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PETadex: A comprehensive database of plastic-degrading enzyme sequence diversity

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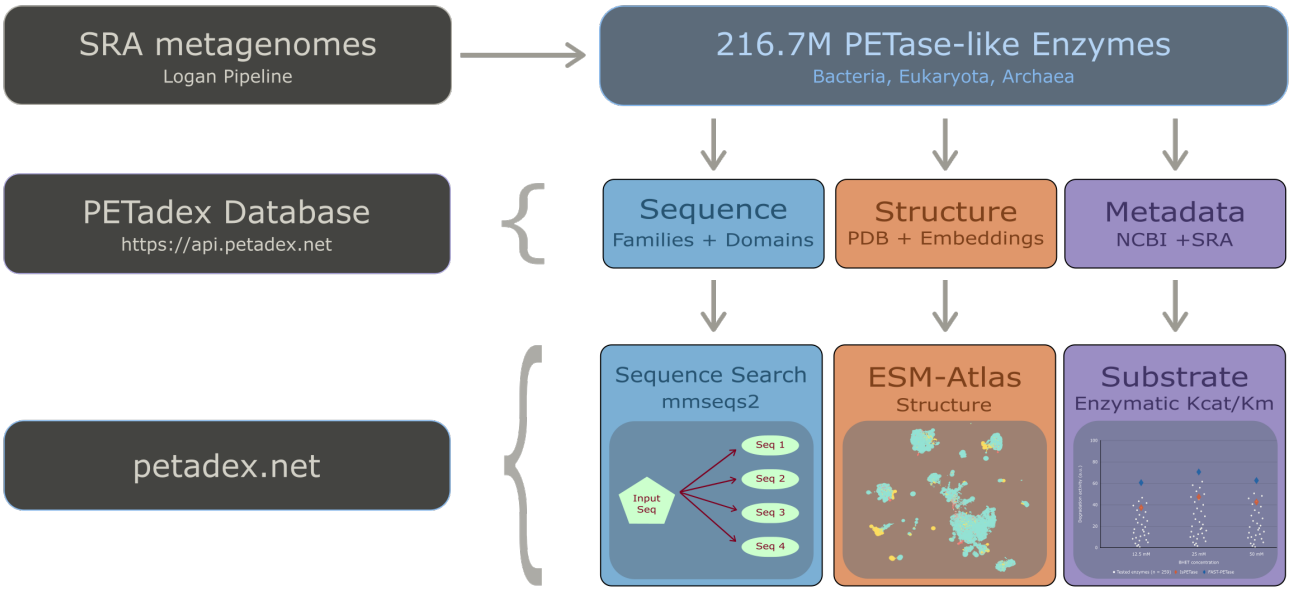
Abstract

Microbial enzymes capable of degrading plastics have been identified across diverse taxa, and this enzymatic environmental diversity represents a rich resource for enzyme discovery and engineering. The vast majority of this diversity, encoded within metagenomic sequencing data, remains uncatalogued and inaccessible. To address this, we present PETadex, a comprehensive database and web platform cataloguing 1.3 billion plastic-degrading enzyme-like sequences identified from the Logan assembly of the Sequence Read Archive, supplemented with 1.05 million sequences from NCBI nr, represented by 216.7 million cluster centroids at 90% sequence identity. The 1.05 million NCBI nr sequences are further organized into 64,730 families through a three-level clustering hierarchy, spanning nearly all major structural classes of known plastic-degrading enzymes. The database integrates functional annotations including InterProScan domain assignments and SignalP signal peptide predictions, NCBI metadata for 2.7 million non-redundant variant sequences, and 259 activity measurements from a standardized BHET hydrolysis assay enabling direct quantitative comparison across enzymes. The web platform provides name-based lookup of catalogued enzymes with associated sequences, structural annotations, metadata, and activity data, as well as MMseqs2-powered sequence similarity search for classifying novel sequences against the PETadex database. An interactive ESM-2 embedding landscape (Atlas) enables visual exploration of enzyme family relationships, capturing structural and functional similarity beyond sequence identity. Applying this landscape revealed biologically coherent clusters containing multi-domain and sulfur-associated architectures that sequence-identity clustering alone does not resolve. PETadex is fully open-source and available at `https://petadex.net`.

Keywords PETase, plastic degradation, metagenomics, enzyme database, protein language model, sequence search

Graphical Abstract



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Introduction

Plastic pollution is a defining environmental challenge of the 21st century. Global plastic production exceeded 400 million tonnes in 2022, and polyethylene terephthalate (PET) accounts for a substantial fraction of this output, driven primarily by beverage bottle and textile fiber manufacturing (Plastics Europe, 2023). PET is highly resistant to natural degradation, persisting in the environment for centuries and accumulating across marine, freshwater, and terrestrial ecosystems (Chamas et al., 2020). The resulting ecological and public health consequences have motivated an urgent search for sustainable remediation strategies, among which biological degradation by microbial enzymes has emerged as a particularly promising approach (Wei and Zimmermann, 2017).

The landmark discovery of the bacterium *Ideonella sakaiensis* and its PET-degrading enzyme (IsPETase) in 2016 demonstrated that nature has already evolved enzymatic machinery capable of depolymerizing PET under mild conditions (Yoshida et al., 2016). Since then, a growing number of PET hydrolases have been identified from diverse microbial taxa, including cutinases, lipases, and esterases with varying degrees of PET-degrading activity (Danso et al., 2019). Engineering efforts have produced thermostable and catalytically enhanced variants with potential industrial applications, most notably FAST-PETase and LCC (Lu et al., 2022; Tournier et al., 2020). These advances have established enzymatic PET degradation as a viable component of circular plastic economy strategies.

Though PET-degrading enzymes have been found across a wide range of microbial taxa, the number that have been experimentally characterized remains small, and the resources available for exploring this sequence diversity are limited in scope. As of April 2026, the PAZy database provides a curated catalogue of 478 experimentally verified plastic-degrading enzymes, serving as the gold standard for literature-validated data (Buchholz et al., 2022). PlasticDB links 329 proteins and 875 species to plastic degradation with associated metadata (Gambarini et al., 2022). While both resources are valuable for their curation depth, they operate at a fundamentally different scale from the vast sequence diversity recoverable from modern metagenomic datasets. The explosion of environmental sequencing through initiatives such as the Sequence Read Archive (SRA) has generated petabases of data in which millions of uncharacterized PETase-like sequences are embedded, yet no existing platform provides systematic access to this diversity (Leinonen et al., 2011).

To address this gap, we developed PETadex, a database and web platform that catalogues 1.3 billion PETase-like protein sequences mined from environmental sequencing data, represented by 216.7 million cluster centroids at 90

Materials and Methods

Data sources and sequence acquisition

The PETadex sequence database was constructed through a multi-stage homology search pipeline originating from the PAZy database of experimentally validated plastic-active enzymes (Buchholz et al., 2022). PAZy sequences were used as queries in a DIAMOND2 search (–very-sensitive) against the NCBI non-redundant protein database, retrieving approximately 2.7 million unique sequences (e-value ¡ 1e-5). These were filtered using hmmscan against Pfam models for known plastic-degrading enzyme domains (e-value ¡ 1e-5, 95% model coverage), yielding a validated set of candidate centroids. We refer to this set of nr-derived sequences as *PETadex-nr*. These centroids were subsequently searched against the Logan planetary-scale assembly of the Sequence Read Archive (27.3 million accessions; December 2023 freeze) (Chikhi et al., 2025) using DIAMOND2 (e-value ¡ 1e-8), with the same HMM filtering applied to the results. We refer to the resulting SRA-derived sequences as *PETadex-Logan*. Together, PETadex-nr and PETadex-Logan comprise 1.3 billion unique PETase-like protein sequences spanning diverse microbial taxa and environmental contexts. Clustering at 90

Sequences were clustered using MMseqs2 (Steinegger and Söding, 2017) at multiple identity thresholds to produce a three-level hierarchy: components (broad groupings at 10

Functional annotation

Functional domain annotations were computed for all family centroids (approximately 65,000 sequences) using InterProScan 5.72-103.0 (Jones et al., 2014), searching against Pfam, TIGRFAM, CDD, SUPERFAMILY, Gene3D, and PANTHER. Signal peptide predictions were generated using SignalP 6.0 (Teufel et al., 2022) on the same centroid set. These centroid-level annotations provide a functional summary for each family, though they should not be assumed to generalize to all member sequences within a cluster. Efforts to expand annotation coverage across the full dataset are ongoing. All annotations are surfaced directly in the web interface, allowing users to browse, filter, and explore domain architectures interactively.

Metadata enrichment

Metadata were retrieved for sequences across both PETadex-nr and PETadex-Logan. For the approximately 2.7 million PETadex-nr sequences, metadata fields include organism, protein definition, taxonomy, journal of origin, collection date, and country. For PETadex-Logan sequences, sample-level metadata from the SRA provides complementary fields such as geographic coordinates, temperature, and pH. To reconcile inconsistencies between the NCBI and SRA metadata schemas, computational standardization was applied to harmonize field formats and vocabularies across sources, improving the dataset's accessibility for large-scale programmatic queries. These metadata are accessible through the MMseqs2 sequence search and the interactive ESM-2 embedding landscape, with dedicated browsing and filtering interfaces under development.

Database architecture and web platform

PETadex is deployed as a three-tier web application (Figure 1). The frontend is a static site built with Gatsby (Gatsby, 2024) and hosted on GitHub Pages. User requests are routed through AWS API Gateway to a serverless Node.js/Express backend deployed on AWS Lambda via the Serverless Framework (Serverless, 2024). The backend exposes

a REST API that supports programmatic access to all database queries, enabling integration with external tools and automated workflows in addition to the web interface. The backend queries a PostgreSQL database hosted on AWS RDS, which stores the core data entities: enzyme sequences, clustering taxonomy (component, family, centroid relationships), the variant dictionary, NCBI/SRA metadata, InterProScan annotations, SignalP predictions, experimental activity data, and precomputed UMAP coordinates. Precomputed materialized views in PostgreSQL enable responsive queries at scale, avoiding expensive joins at request time. Phylogenetic trees and other precomputed assets are stored in Amazon S3 for asynchronous retrieval by the client. Sequence search functionality is described in Section 2.6, and the ESM-2 embedding landscape in Section 2.5. The entire PETadex codebase is open-source and publicly available at `https://github.com/petadex/petadex.io`.

ESM-2 embedding landscape

Protein language model embeddings were computed for all 64,730 PETadex-nr family centroids using ESM-2 (`esm2_t30_150M_UR50D`) (Lin et al., 2023), selected as a representative middle-ground between computational efficiency and embedding quality within the ESM-2 model family. Sequences exceeding the model's 1,022-residue context window were truncated, and non-standard IUPAC amino acid codes were remapped to ESM-2-compatible equivalents. Mean-pooled residue-level representations were extracted from the final transformer layer (layer 30), excluding beginning- and end-of-sequence tokens, to produce fixed-length 640-dimensional embeddings for each centroid. Embeddings were generated in batches of 64 on an NVIDIA T4 GPU (AWS EC2 g4dn.xlarge instance) in approximately 57 minutes. Dimensionality reduction was performed using UMAP (McInnes et al., 2020). The resulting atlas was validated by confirming that known protein domain families and taxonomic groupings clustered coherently in the projected space. Coordinates are stored in PostgreSQL alongside family-level metadata and rendered on the frontend as an interactive scatter plot using deck.gl's ScatterplotLayer over a WebGL canvas.

Sequence search

PETadex provides protein sequence search powered by MMseqs2 (Steinegger and Söding, 2017), executed via a dedicated AWS Lambda function. Users submit a FASTA-formatted protein sequence through the web interface with a configurable maximum number of results. The Lambda function runs MMseqs2 with default parameters (sensitivity 5.7, e-value threshold 0.001) against the PETadex sequence database and returns, for each hit: accession, organism, percent identity, E-value, bit score, query coverage, alignment length, and query/target coordinate ranges. Results are further enriched with database-derived fields including family classification, component assignment, and phylogenetic tree availability. Identical queries are cached using deterministic deduplication to avoid redundant computation. Response metadata includes the database size, search execution time, and total number of hits returned.

Substrate activity comparison

Experimental BHET hydrolysis data displayed in PETadex are sourced from the enzyme validation experiments reported in Chikhi *et al.* (Chikhi et al., 2025). In that study, candidate PETase genes were synthesized, expressed as yeast surface-displayed enzymes, and robotically pinned onto YPD agar plates containing BHET at three concentrations (12.5, 25, and 50 mM). Plates were imaged at five timepoints (8, 24, 48, 72, and 96 h) before and after colony removal, and hydrolytic activity was quantified as background-normalized median pixel intensity under each colony using a custom image analysis pipeline. PETadex serves these activity measurements through a dedicated substrate comparison interface (`/substrate`), displaying per-gene intensity profiles across timepoints and BHET concentrations (Figure 4). A degradation activity metric is computed server-side for each gene and substrate pair as the difference between the peak intensity value and the subsequent minimum intensity value across the timeseries. Four quality flags are assigned where applicable: `peak_at_end` (peak falls at the final timepoint; activity set to null), `peak_at_start` (peak at the first timepoint), `single_degradation_point` (only one timepoint separates peak from minimum), and `negative_activity` (no degradation signal detected). These flags are surfaced in the web interface and included in downloaded data to allow users to apply their own quality thresholds.

Results

Database scale and composition

PETadex contains 1.3 billion PETase-like protein sequences, represented by 216.7 million cluster centroids at 90% sequence identity, organized into approximately 65,000 families and 43 components through the three-level clustering hierarchy described above. The non-redundant variant dictionary comprises 2.7 million individual sequences with associated NCBI metadata, providing the variant-level resolution needed for taxonomic and environmental analysis. Family size follows a heavy-tailed distribution, with a small number of large families containing thousands of variants and a long tail of small families with fewer than ten members (Figure 2C). The database also includes 259 standardized BHET hydrolysis activity measurements sourced from Chikhi *et al.* (Chikhi et al., 2025), providing experimentally grounded activity data for a subset of catalogued enzymes.

Taxonomic analysis of the metadata-annotated fraction reveals that bacteria dominate the database (89.5

Temporal analysis of the 38.2% of sequences with recorded collection dates shows a marked increase in PETase-like sequence deposition from 2015 onward, coinciding with the characterization of IsPETase by Yoshida *et al.* (Yoshida et al., 2016) (Figure 2D). This pattern likely reflects both increased research interest in PET-degrading enzymes and the broader expansion of environmental sequencing over the same period, making it difficult to attribute the trend to either factor alone. Metadata field completeness is high for core identifiers and taxonomy (over 99.7%) but sparse for temporal and geographic annotations (Figure 2E). Country of origin is absent from all

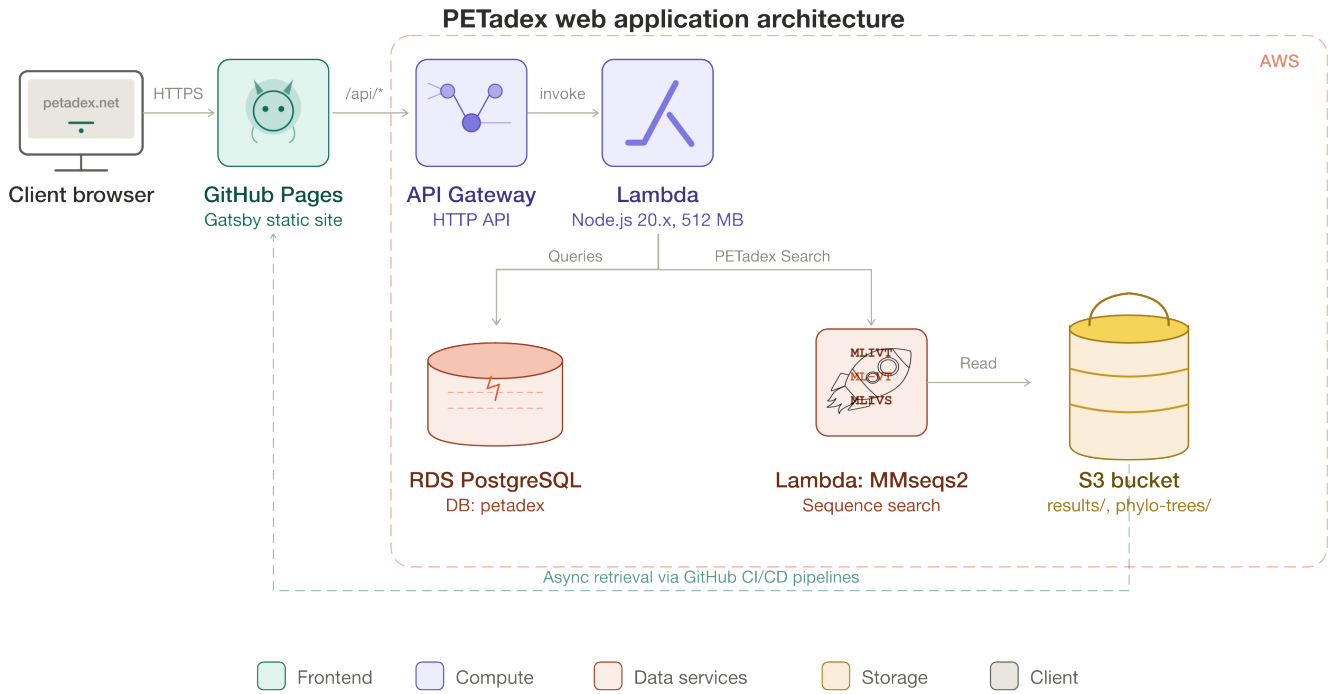

Figure 1 Architecture of the PETadex web platform. A Gatsby static site (petadex.net), hosted on GitHub Pages, routes API requests through AWS API Gateway to a serverless Node.js 20.x backend (AWS Lambda), which queries a PostgreSQL database (AWS RDS) and invokes a dedicated Lambda function for MMseqs2 sequence search. The MMseqs2 Lambda reads precomputed results and phylogenetic trees from an Amazon S3 bucket. Asynchronous data retrieval is managed via GitHub CI/CD pipelines.

records in the current release, limiting the ability to perform spatial analyses of PETase diversity.

Biological signal from the embedding landscape

The ESM-2 Atlas provides a structure-aware view of PETase family relationships that complements the sequence-identity-based clustering (Figure 3). During exploration of the embedding landscape, we observed that a number of families cluster together in embedding space despite being assigned to distinct families by MMseqs2 clustering. Examination of these families revealed multi-domain architectures in which a PETase-like catalytic domain is fused to one or more additional enzymatic domains, producing so-called "cocktail enzymes." Because the sequence-identity clustering operates on full-length sequences, these chimeric proteins are assigned to families that reflect their overall sequence composition rather than their individual domain content. The ESM-2 embeddings, which capture structural and functional features that extend beyond primary sequence similarity, group these multi-domain proteins together based on shared domain-level relationships, effectively flagging families that may benefit from domain-level decomposition or reclassification.

Cross-referencing the embedding coordinates of these cocktail enzyme families with NCBI environmental metadata revealed an enrichment of sequences originating from sulfur-rich, aerobic environments, including volcanic and hydrothermal contexts (Figure 3). The co-occurrence of PETase-like domains in bacteria from sulfur-associated niches raises the possibility of a non-plastic ecological role for these enzymes, or alternatively, convergent recruitment of hydrolase domains in organisms adapted to chemically extreme environments. We emphasize that this association is preliminary and correlative; systematic quantification of this enrichment and deeper characterization of these cocktail enzyme families, including domain boundary analysis, phylogenetic reconstruction, and experimental validation, is ongoing.

Substrate activity comparison

The substrate activity comparison interface enables direct quantitative comparison of BHET hydrolysis activity across the 259 enzymes with standardized experimental data (Figure 4). Summary statistics are displayed for each BHET concentration (12.5, 25, and 50 mM), and a scatter plot of mean activity intensity across concentration pairs reveals a strong positive correlation, indicating that enzymes with high activity at lower BHET concentrations generally maintain elevated activity at higher concentrations. Notably, a subset of enzymes exhibit disproportionately high activity at 25 mM relative to 12.5 mM, suggesting concentration-dependent activation or substrate saturation effects that may warrant further biochemical investigation. The interface supports filtering by timepoint and

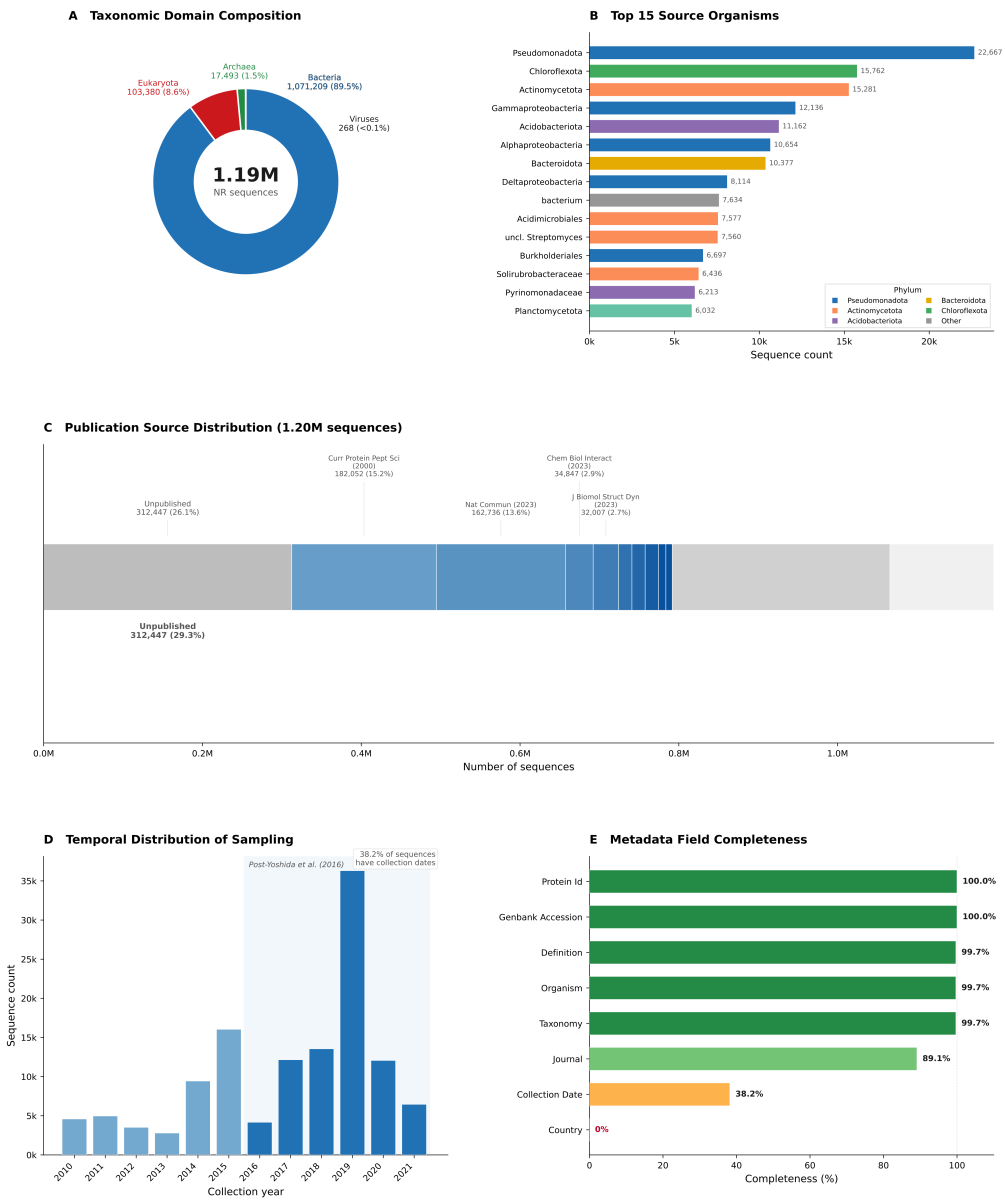

Figure 2 Metadata landscape of the PETadex non-redundant sequence database. (A) Taxonomic domain composition of 1,196,644 annotated sequences, dominated by Bacteria (89.5%) with representation across Eukaryota, Archaea, and Viruses. (B) Top 15 source organisms by sequence count, colored by phylum. (C) Family size distribution across the database. (D) Temporal distribution of sample collection years, showing increased deposition following the characterization of IsPETase (Yoshida et al., 2016). (E) Metadata field completeness across all records.

concentration, allowing users to identify candidate enzymes with desirable activity profiles for downstream characterization or engineering.

Querying IsPETase against PETadex

To illustrate the platform's utility for sequence-level exploration, we queried the canonical *Ideonella sakaiensis* PETase (IsPETase, 290 amino acids) against the PETadex database (Figure 5). The search returned 50 hits spanning a wide range of sequence identity, from a perfect self-match (100% identity, family 20180) down to remote homologs at 36.1% identity. The results mapped to at least seven distinct PETadex families, indicating that PETase-like sequence space is fragmented across multiple evolutionarily distinct enzyme groups rather than forming a single monolithic cluster.

Several experimentally validated PET and polyester hydrolases appeared at varying evolutionary distances, including the *Acidovorax delafieldii* poly(butylene succinate-co-adipate) depolymerase (78.9% identity, family 5885), the *Thermobifida cellulosilytica* cutinase (52.7% identity), and the *Halopseudomonas aestusnigri* hydrolase (51.5% identity). A PDB structural entry (7DZT_A, 82.5% identity) also appeared among the hits, linking the search directly to available crystallographic data. The full set of search results can be explored interactively at `https://petadex.net/results/?job=regen_example_ispetase`.

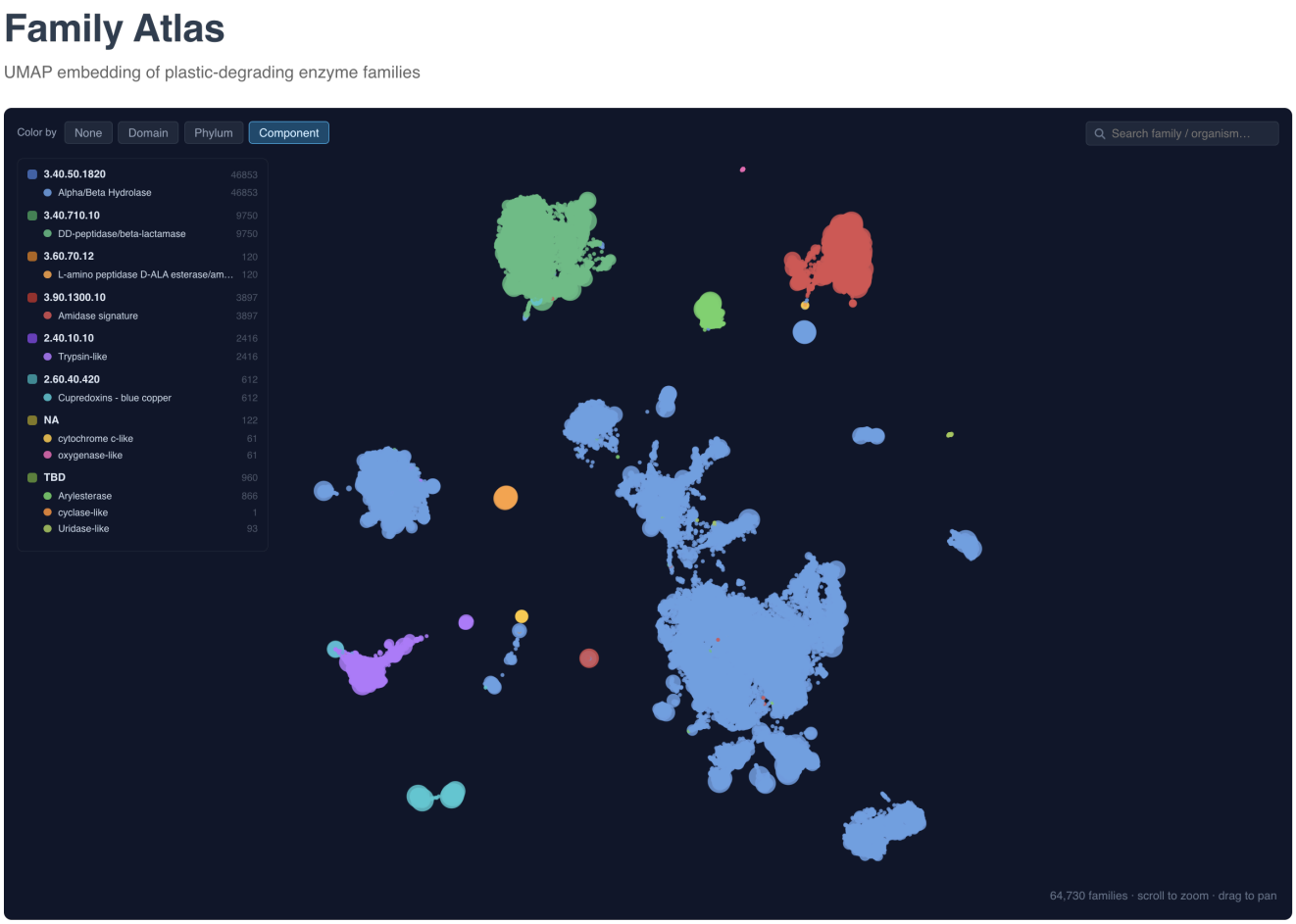

Figure 3 ESM-2 embedding landscape (Atlas) of PETadex family centroids. UMAP projection of ESM-2 embeddings computed for approximately 65,000 family centroids. Each point represents a PETadex family, with spatial proximity reflecting structural and functional similarity as captured by the protein language model. The Atlas is available as an interactive visualization at `https://petadex.net/atlas`.

Discussion

PETadex represents a fundamentally different kind of resource from existing PETase databases. PAZy, the current gold standard for curated PET-degrading enzymes, catalogues 478 entries with activity data aggregated from heterogeneous assays across the published literature (Buchholz et al., 2022). While invaluable for its curation depth, PAZy's activity measurements are drawn from diverse experimental conditions, substrates, and reporting standards, which limits direct quantitative comparison across enzymes. PlasticDB provides a complementary resource linking 329 proteins and 875 species to plastic degradation with associated metadata (Gambarini et al., 2022). Both databases serve as curated literature resources. PETadex, in contrast, is a discovery platform designed to make the full predicted diversity of PET-degrading enzymes from metagenomic data navigable and searchable at a scale of 1.3 billion sequences represented by 216.7 million centroids.

This difference in scope is not merely quantitative. Curated databases, by design, index enzymes that have been individually cloned, expressed, and characterized in the literature, which biases their coverage toward a small number of well-studied enzyme families and culturable organisms. PETadex derives its sequences from an exhaustive search of metagenomic and environmental sequencing data in the SRA, capturing PETase-like diversity from uncultured lineages, undersampled biomes, and taxonomic groups that have never been the focus of directed PET degradation studies. The result is a database that reflects the full predicted sequence landscape of PET hydrolases rather than the subset visible through the lens of traditional enzyme characterization.

A notable feature of the experimental data within PETadex is its internal consistency. The 259 activity datapoints in the database derive from a single standardized BHET hydrolysis assay performed under uniform conditions, as described in Chikhi *et al.* (Chikhi et al., 2025). Because all measurements were generated using the same assay protocol, substrates, and conditions, they enable direct quantitative comparison across enzymes in a way that data aggregated from heterogeneous literature sources cannot support. The substrate activity comparison interface (Figure 4) surfaces these data interactively, and the sequence-to-activity join path in PETadex links computational sequence data directly to these standardized measurements, providing a foundation for structure-activity analyses and machine learning applications. Integration of additional published activity data from PAZy and the primary

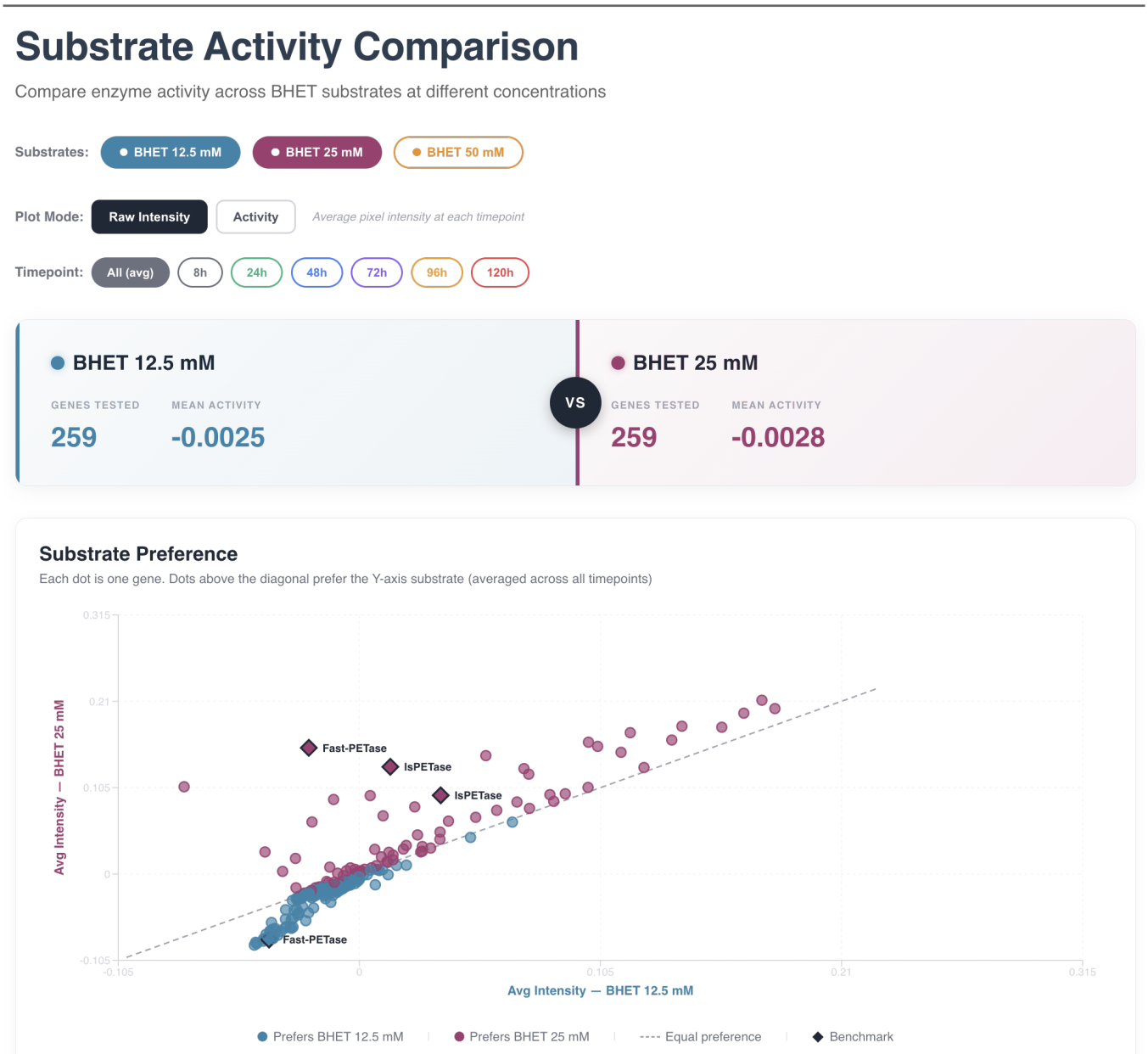

Figure 4 Substrate activity comparison interface in PETadex. The interface displays BHET hydrolysis activity data for 259 enzymes under standardized assay conditions. Top: summary cards showing the number of genes tested and mean activity at each BHET concentration. Bottom: scatter plot comparing mean activity intensity between BHET 12.5 mM and 25 mM conditions, with points colored by concentration. The interface allows filtering by substrate concentration and timepoint to identify enzymes with desirable activity profiles.

literature is planned and will further expand the experimentally grounded fraction of the database.

The cocktail enzyme finding described in Section 3.2 illustrates the value of combining sequence-identity clustering with structure-aware embeddings. These two approaches capture complementary aspects of protein relationships, and by providing both views on the same data, PETadex enables researchers to identify patterns that would not emerge from either perspective alone.

Limitations

Several limitations of PETadex should be noted. The dataset is derived entirely from homology search against known plastic-degrading enzyme domains, and therefore cannot capture PET-degrading enzymes with novel folds or domains not represented in the PAZy seed set. The taxonomic and environmental composition of the database reflects the biases inherent in the SRA, which overrepresents certain environments, sequencing platforms, and geographic regions. Metadata completeness remains uneven: geographic annotations are largely absent, and temporal metadata are available for only 38.2% of sequences. Finally, the vast majority of sequences in PETadex are computationally predicted homologs whose PET-degrading

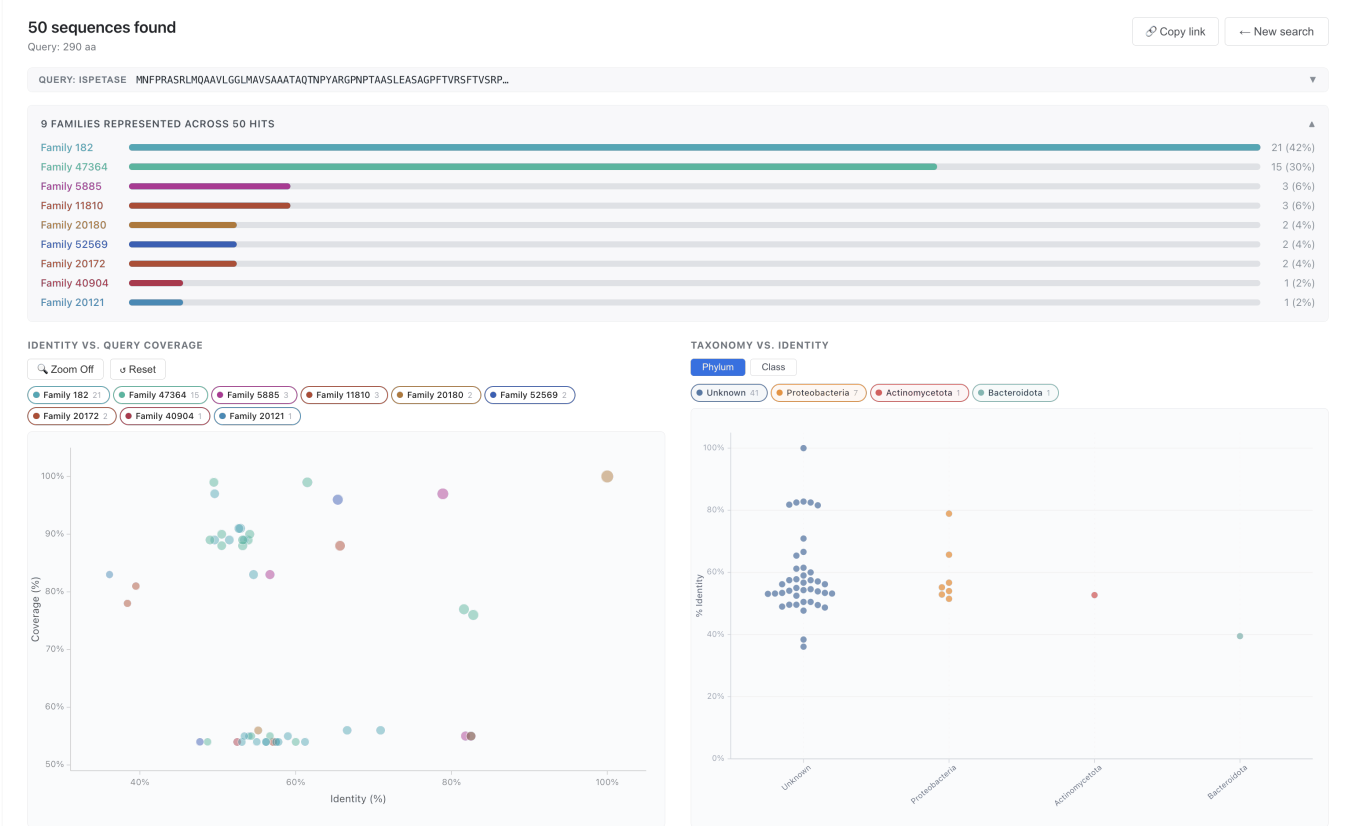

Figure 5 MMseqs2 search results for IsPETase queried against PETadex. Beeswarm plot showing the distribution of 50 hits across percent identity, colored by PETadex family assignment. Experimentally characterized enzymes are labeled. The search recovers hits from at least seven distinct families spanning diverse bacterial lineages.

activity has not been experimentally verified. The database should therefore be understood as a resource for hypothesis generation and candidate prioritization rather than a catalogue of confirmed PETases.

Future directions

Several extensions of PETadex are planned or underway. We are developing automated pipelines to scrape, normalize, and integrate published activity data from PAZy, the primary literature, and supplementary tables, which will expand the experimentally grounded fraction of the database and enable more comprehensive structure-activity mapping. Similarly, integration of experimental activity annotations into the sequence search output is under development, which will allow users to identify experimentally tested homologs directly from search results. We are also developing PETai, a machine learning pipeline that uses ESM-2 embeddings to predict PETase activity from sequence features. A proof-of-concept has been implemented, and the ESM-2 embeddings used for training are publicly available as part of the open-source PETadex codebase. For structural annotation, we plan to cross-reference family centroids against the AlphaFold Protein Structure Database and selectively generate predictions for representative centroids and proteins of particular interest using AlphaFold2 (Jumper et al., 2021) or ESMFold (Lin et al., 2023), as full-scale folding of all 65,000 centroids is not currently feasible. Longer-term goals include community submission tools that would allow researchers to deposit new sequences and activity measurements directly into the platform.

Data Availability

PETadex is freely available at `https://petadex.net`. The database can be queried through the web interface or programmatically via the REST API. Source code for the web platform is available at `https://github.com/petadex/petadex.io` and the sequence search infrastructure at `https://github.com/petadex/petadex-sequence-search`. The underlying sequence data and clustering are described in Chikhi *et al.* (Chikhi et al., 2025).

Author Contributions

D.Z. developed the PETadex web platform, database architecture, serverless backend, sequence search infrastructure, ESM-2 embedding landscape, data visualizations, and NCBI metadata enrichment pipeline. W.A. contributed to frontend web elements. T.Q. assisted with the database backend. A.B. supervised the project. The sequence discovery pipeline and BHET activity assay data were generated by Chikhi *et al.* (Chikhi et al., 2025). D.Z. wrote the manuscript with supervision from A.B.

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